

## Bias-variance (almost guaranteed)

$$\mathbb{E}_{D,\varepsilon}[(y_0 - \hat{f}(x_0))^2] = \underbrace{(f - \mathbb{E}\hat{f})^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f})}_{\text{var}} + \underbrace{\varepsilon^2}_{\text{irred.}}$$

**Derive:**  $y_0 = f + \varepsilon$ , expand  $(f - \hat{f} + \varepsilon)^2$ . Cross  $2(f - \hat{f})\varepsilon$  dies by  $\mathbb{E}\varepsilon = 0$  (not independence);  $\mathbb{E}\varepsilon^2 = \sigma^2$ . Then  $\pm \mathbb{E}\hat{f}$ :  $(f - \mathbb{E}\hat{f}) + (\mathbb{E}\hat{f} - f)$ , cross dies by LIE.

**Exact identity, not a law.** Prof rejects “trade-off” because (i) regularisation cuts variance at small bias cost (bias is *squared*); (ii) over-param regime: past interpolation peak ( $p > n$ ) both shrink (**double descent**). Mechanism: among  $\infty$  zero-train solutions, SGD/OLS<sup>+</sup> pick the **min-norm interpolator**  $\Rightarrow$  implicit  $L^2$ .

**Flex. direction:** train MSE  $\downarrow$ ; test MSE  $U$ ; bias<sup>2</sup>  $\downarrow$ ; var $\uparrow$ ;  $\sigma^2$  flat. KNN flex =  $1/K$ .

**T/F traps (all F):** bias-var only for prediction; test MSE  $\rightarrow 0$  as  $n \rightarrow \infty$ ; lower bias  $\Rightarrow$  better preds; high  $\sigma^2 \Rightarrow$  want flex.

Classif analogue: Bayes error =  $1 - \mathbb{E}[\max_k P(Y=k|X)]$  plays  $\sigma^2$ .

## Random vectors / MVN

$Z = CX \Rightarrow \mathbb{E}Z = C\mu$ ,  $\text{Cov}(Z) = C\Sigma C^T$  (transpose on *right*).  $\rho_{ij} = \sigma_{ij}/(\sigma_i\sigma_j)$  (sqrt trap:  $\Sigma_{ii} = \sigma_i^2$ ). Zero cov  $\neq$  indep *except under joint normality*. MVN density  $\propto \exp(-\frac{1}{2}D_M^2)$ ,  $D_M^2 = (x - \mu)^T \Sigma^{-1}(x - \mu) \sim \chi_p^2$ . Singular  $\Sigma \Rightarrow$  LDA/MVN blows up.

$\Sigma$ -to-contour: equal var  $\rho = 0$  circle; unequal  $\rho = 0$  axis-aligned (long axis = larger var);  $\rho > 0$  tilt  $\nearrow$ ;  $\rho < 0$  tilt  $\nwarrow$ .

## Linear regression: interpretation

**Output table:**  $\hat{\beta} \pm 2$  SE is 95% CI; sig. at 5% iff  $|t| \geq 2$ ; residual df =  $n - p - 1$  (sanity-check param count vs table rows).

**Interaction trap.**  $y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ . “Main effect”  $\beta_2$  is the Z-effect *only at*  $X = 0$ . Per-group lines:  $Z = 0 : y = \beta_0 + \beta_1 X$ ;  $Z = 1 : y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X$ . **State your coding.** Hierarchical: never drop a main effect while keeping its interaction. Non-sig main in interaction model  $\nRightarrow$  drop.

**Dummies.**  $K$  levels  $\rightarrow K-1$  dummies; coefs are *differences from reference*; ref-flip  $\Rightarrow$  all signs flip, predictions unchanged. Never 0/1/2 ordinal coding.

**Stat vs practical sig.**  $\text{SE}(\hat{\beta}_j) \propto 1/\sqrt{n}$ , so for big  $n$  any effect is significant. Quote effect size +  $p$ . “Linear in  $\beta$ , not in  $X$ .”

**What  $p$  is NOT:**  $p \neq \Pr(H_0|\text{data})$ ;  $p > 0.05 \nRightarrow H_0$  true; “fail to reject”  $\neq$  accept.

**Independence is load-bearing.** 5 assumptions; (4)+(5) violations ruin SE/CI/ $p$  (time-series, spatial, clustered). Gaussian/zero-mean/homosced. robust.

## LR diagnostic plots

**Resid vs fitted:** curvature $\Rightarrow$ nonlinearity; fanning $\Rightarrow$ heterosc.; clusters $\Rightarrow$ indep break.

**QQ** (prof: “would put on test”): straight $\Rightarrow$ Gaussian; S $\Rightarrow$ tails; banana $\Rightarrow$ skew; kinks $\Rightarrow$ outliers.

**Scale-location:**  $\sqrt{|\hat{r}|}$  vs fitted; trend $\Rightarrow$ var scales with mean.

**Resid vs leverage:** danger = high  $h_{ii}$  and large  $|\hat{r}|$ . Cook’s  $D_i \propto \hat{r}_i^2 h_{ii}/(1 - h_{ii})$ ; flag  $> 0.5$ , def.  $> 1$ .

**Lev vs infl (fat kid on seesaw):**  $h_{ii} \in [1/n, 1]$  depends on  $X$  alone; influence = leverage  $\times$  residual.  $\sum h_{ii} = p + 1$ . Std. resid  $\hat{r}_i = e_i/(\hat{\sigma}\sqrt{1 - h_{ii}})$ .  $\hat{\sigma}^2 = \text{RSS}/(n - p - 1)$ .

**Collinearity sx:** high SEs, indiv  $t$  insig BUT joint  $F$  sig; signs flip across resamples; preds OK, interp broken. Fix: drop / ridge / PCR / combine.

**CI vs PI:**  $\hat{y} \pm t\hat{\sigma}\sqrt{x_0^T(X^T X)^{-1}x_0}$  (mean) vs  $\sqrt{1 + \cdot}$  (future). *PI always wider*. Freq. interp: “over many samples, 95% of CIs cover  $\beta$ ”—a single CI either does or not.

“Fancy correlations, not causal.” Errors  $\varepsilon_i$  unobservable; residuals  $e_i = y_i - \hat{y}_i$  predict them.

## Logistic & odds (every exam)

$\eta = \beta_0 + \beta^T x$ ,  $p = \sigma(\eta) = 1/(1 + e^{-\eta})$ , logit  $p = \eta$ . Fit by **MLE / Newton-Raphson** (no closed form).  $\widehat{\text{Var}}(\hat{\beta}) = (X^T W X)^{-1}$ ,  $W = \text{diag}(p_i(1 - p_i))$ . odds =  $p/(1 - p)$ ,  $p$  = odds/(1 + odds).  $\Delta x_j = 1 \Rightarrow$  odds  $\times e^{\beta_j}$  (**NOT probability change**).  $\Delta x_j = k \Rightarrow e^{k\beta_j}$ . Interaction:  $\text{OR}_j = e^{\beta_j + \beta_{jk}x_k}$  (depends on  $x_k$ ). **Ref relabel**  $\Rightarrow$  sign flip on that dummy; preds unchanged. Threshold inversion:  $x_{0j} = \lfloor \logit(p^*) - \beta_0 - \sum_{k \neq j} \beta_k x_{0k} \rfloor / \beta_j$ .

## LDA / QDA (often the mathy Q)

$X|Y = k \sim N(\mu_k, \Sigma)$ , prior  $\pi_k$ . LDA  $\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$  (linear:  $x^T \Sigma^{-1} x$  cancels across  $k$ ).

**1D 2-class boundary** ( $\pi$  unequal):  $x = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_1 - \mu_2} \log \frac{\pi_2}{\pi_1}$ .

**Posterior:**  $P(Y = k|x) = e^{\delta_k(x)} / \sum_\ell e^{\delta_\ell(x)}$  (softmax of  $\delta$ 's; LDA/QDA/NB).

**Pooled cov:**  $\hat{\Sigma} = \frac{1}{n-K} \sum_k (n_k - 1) S_k$ .

**QDA** (drop pooling):  $\delta_k = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) - \frac{1}{2} \log |\Sigma_k| + \log \pi_k$ . Quadratic survives because  $x^T \Sigma_k^{-1} x$  no longer cancels. Don't forget  $-\frac{1}{2} \log |\Sigma_k|$ .

**Boundary derivation (LDA 2D):** (1)  $\Sigma^{-1}$ ; (2)  $v = \Sigma^{-1}(\mu_A - \mu_B)$ ; (3)  $c = -\frac{1}{2} \mu_A^T \Sigma^{-1} \mu_A + \frac{1}{2} \mu_B^T \Sigma^{-1} \mu_B + \log(\pi_A/\pi_B)$ ; (4)  $v^T x + c = 0$ . Worked:  $\mu_A = (1, 1)$ ,  $\mu_B = (3, 3)$ ,  $\Sigma = 2I$ ,  $\pi$  eq.  $\Rightarrow x_2 = 4 - x_1$ .

**Prior shift:**  $\pi_k \uparrow \Rightarrow$  boundary moves *away from* class  $k$ . Normal to boundary is  $\Sigma^{-1}(\mu_A - \mu_B)$ , *not*  $\mu_A - \mu_B$  unless  $\Sigma \propto I$ .

**Compare:** Logistic (linear, no  $X$ -assumption,  $p+1$  par); LDA (linear, Gaussian pooled,  $p(p+1)/2$ ); QDA (quadratic,  $K \cdot p(p+1)/2$ ); Naive Bayes (diag  $\Sigma_k$ ,  $2pK$ ); KNN (nonpar, dies in high  $p$ ).

**Multi-class logistic, probit, Fisher LDA:** OUT.

## Confusion / ROC / AUC

|       |             |             |                              |
|-------|-------------|-------------|------------------------------|
|       | $\hat{y}=0$ | $\hat{y}=1$ |                              |
| $y=0$ | TN          | FP          | sens=TPR= $\frac{TP}{TP+FN}$ |
| $y=1$ | FN          | TP          |                              |

spec =  $\frac{TN}{TN+FP}$ ; FPR =  $1 - \text{spec}$ ; acc =  $1 - \text{err}$ .

**Mnemonic:** Sens sniffs  $\uparrow$ , Spec spares  $-$ .

Threshold  $\downarrow$ : sens $\uparrow$ , spec $\downarrow$ , op-point  $\nearrow$  on ROC.

**Imbalance:** always-majority classifier has acc =  $1 - \text{min-rate}$ , sens = 0.

ROC:  $y = \text{TPR}$  vs  $x = \text{FPR}$ .  $t = 1 \rightarrow (0, 0)$ ,  $t = 0 \rightarrow (1, 1)$ , diag = chance. **AUC** =  $\Pr(\hat{p}(X^+) > \hat{p}(X^-))$ ; invariant to monotone  $\hat{p}$  transforms; AUC  $< 0.5 \Rightarrow$  flip. Same AUC  $\neq$  same shape.

## Resampling

**LOOCV OLS shortcut (1 fit):**  $\text{CV}_{(n)} = \frac{1}{n} \sum_i \left( \frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$ . Works for any linear smoother ( $S$  replaces  $H$ ).

**Bias/var of CV:**

|                         | bias | var | cost |
|-------------------------|------|-----|------|
| val 50/50               | hi   | hi  | 1    |
| LOOCV                   | lo   | hi  | $n$  |
| $k$ -fold ( $k=5, 10$ ) | med  | lo  | $k$  |

LOOCV high var because train sets share  $n-2$  obs  $\Rightarrow$  folds correlated.

**1-SE rule** (prof's preferred):  $\theta^* = \text{simplest}$  model with  $\text{CV} \leq \text{CV}(\hat{\theta}) + \hat{\text{SE}}$ ,  $\hat{\text{SE}}$  = sample SD of per-fold MSEs /  $\sqrt{k}$ . SE “not quite valid” (same folds for select+SE).

**Wrong-way CV** (high-impact trap): any step that touches  $y$  (corr filter, variance filter, supervised PCA) *must live inside each fold*. Filter outside  $\Rightarrow$  CV  $\rightarrow$  optimistic LIE. Nested CV: inner=select, outer=assess. Train/val/test = 3 jobs.

**Indep trap:** naive  $k$ -fold breaks under spatial/temporal autocorrelation; fold whole blocks.

**Bootstrap:** draw  $n$  rows *with replacement* (always size  $n$ ),  $B=10^3$ – $10^4$  resamples;  $\hat{\text{SE}}$  = SD over  $\{\hat{\theta}_b^*\}$ ; **percentile 95% CI** =  $(q_{.025}^*, q_{.975}^*)$ .

$P(\text{obs OOB}) = (1 - 1/n)^n \rightarrow 1/e \approx 0.368 \Rightarrow \sim 37\%$  OOB,  $\sim 63\%$  in-bag. Each obs OOB for  $\sim B/3$  trees  $\Rightarrow$  “free CV.”

## Regularisation & subset selection

**Ridge:**  $\min \|y - X\beta\|^2 + \lambda \sum \beta_j^2$ ;  $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$  (always invertible). Bias  $-\lambda(X^T X + \lambda I)^{-1} \beta$ .  $\text{df} = \sum_j d_j^2/(d_j^2 + \lambda)$ .

**Lasso:**  $+\lambda \sum |\beta_j|$ . *Sparse* via L1 corners on axes; active set  $\leq \min(n, p)$ .

**Elastic net:** ridge + lasso. Intercept never penalised; **standardise**  $X$  (penalty scale-sensitive); don't standardise dummies.

**Picture:** L2 ball (circle)  $\rightarrow$  shrinks, never 0 (“socialist”); L1 ball (diamond, corners on axes)  $\rightarrow$  exact zeros (“capitalist”). Trace plot: lasso paths touch 0 and stay; ridge paths asymptote.  $\lambda \uparrow$ : flex $\downarrow$ , bias $\uparrow$ , var $\downarrow$ .  $\lambda=0$ : OLS.  $\lambda \rightarrow \infty$ : null (ridge asymptotic; lasso reaches all-zero at finite  $\lambda$ ).

**Subset counts:** best =  $2^p$ ; forward/backward =  $1 + p(p+1)/2$ . **Backward needs**  $n > p$ ; forward survives  $p > n$ .

**High-dim** ( $p > n$ ) **survival:** OLS, backward BREAK. Forward, Ridge, Lasso, EN, PCR, PLS OK. “Lasso requires  $p < n$ ” = FALSE (past-exam trap).

**CV > AIC/BIC/ $C_p$ :** latter depend on  $\hat{\sigma}^2$ +distributional assumptions, break at  $p \approx n$ . (Algebra OUT.)

**PCR:** std  $\rightarrow$  PCA  $\rightarrow$  regress on top  $M$  PCs (CV-chosen)  $\rightarrow$  map back  $\hat{\beta}_j = \sum_m \theta_m \phi_{jm}$ .  $\approx$  discretised ridge (hard truncation vs smooth shrink). *Unsupervised*  $\Rightarrow$  no guarantee top PCs are predictive; PLS fixes via  $\max \text{Cov}(X\phi, Y)$ . No var selection.

Refit-on-active-set: lasso to select, OLS on survivors to debias.

## Splines / GAM

**DOF table** (intercept-counting trap):

|                          | incl. int.             | excl.                  |
|--------------------------|------------------------|------------------------|
| poly deg $d$             | $d+1$                  | $d$                    |
| step, $K$ cuts           | $K+1$                  | $K$                    |
| cubic spline, $K$ knots  | $K+4$                  | $K+3$                  |
| natural cubic, $K$ knots | $K+2$                  | $K+1$                  |
| smoothing spline         | $\text{tr}(S_\lambda)$ | $\text{tr}(S_\lambda)$ |

Truncated power cubic basis:  $x, x^2, x^3, (x - c_*)^3_+$ . Continuity through 2nd deriv; 3rd jumps. Natural: linear past boundaries (2 boundary constraints each end = 4 constraints, but  $K+2$  params).

**Smoothing spline:**  $\min \sum (y_i - g(x_i))^2 + \lambda \int g''^2$  ( $g''$ , **squared**;  $g'$  is WRONG).  $\hat{y} = S_\lambda y$  linear smoother;  $\text{df}_\lambda = \text{tr}(S_\lambda)$  non-integer.

$\lambda \uparrow \Rightarrow$  **smoother / less wiggly / df $\downarrow$** . Opposite direction from polynomial degree (T/F bait).  $\lambda=0$ : interp;  $\lambda \rightarrow \infty$ : OLS line.

**GAM:**  $y = \beta_0 + \sum_j f_j(x_j)$ . *No interactions* unless added explicitly.  $\text{df}$  = sum of component  $\text{df}$  (don't double-count intercept). Panels are partial effects on  $\eta$  scale (log-odds for logistic-GAM), not predictions. LOESS span  $\uparrow$  = smoother; dies  $p > 3-4$  (KNN-cousin curse).

## Trees & ensembles

**Cost-complexity prune:**  $C_\alpha(T) = Q(T) + \alpha|T|$ ,  $|T|$  = #terminal nodes. Grow fully, prune backward (weakest-link); CV picks  $\alpha$ . Never early-stop.

**Splits:** regression  $\min_{j,s} \sum_{R_1} (y - \bar{y}_{R_1})^2 + \sum_{R_2} (\cdot)^2$  (*sum*, not mean). Classification impurities: Gini  $G = \sum \hat{p}_{jk}(1 - \hat{p}_{jk})$ ; entropy  $D = -\sum \hat{p} \log \hat{p}$ ; misclass  $E = 1 - \max_k \hat{p}$ .

**Asymmetry (prof flag):** train on Gini/entropy (sensitive to purity), *prune on misclassification*.

**Categorical  $q$  levels:**  $2^{q-1} - 1$  binary splits; binary outcome trick: order by  $\hat{p}_1$ , treat ordinal ( $q-1$  cuts).

**Bagging var (mathy candidate):**  $\text{Var}(\frac{1}{B} \sum f_b) = \rho \sigma^2 + \frac{1-B}{B} \sigma^2 \xrightarrow{B \rightarrow \infty} \rho \sigma^2$ .

Floor at  $\rho \sigma^2 \Rightarrow$  **RF decorrelates** via  $m < p$  random predictors per split. Bagging  $m=p$ ; **RF defaults**  $m=\sqrt{p}$  (classif),  $p/3$  (regr).  $B$  is *not* a tuning param (just “enough”). RF trees grown unpruned.

**VI:** impurity-based (biased to hi-cardinality) vs **permutation OOB** (prof prefers).

**Boosting one-line:** sequential weak learners fit **negative gradient** of loss (squared err  $\equiv$  residuals).  $f_M(x) = \sum_m \nu T(x; \Theta_m)$ . Attacks **bias**.

$\nu$  (lr, 0.01–0.1)  $\downarrow$  needs  $M \uparrow$   
 $M$  trees  $\uparrow$  U:  $\downarrow$  bias then  $\uparrow$  var  
 $J$  leaves (4–8)  $\downarrow$   $J$ –1-way interactions

Boosting *can* overfit in  $M$  (RF cannot).  $M$ – $\nu$  couple: fix  $\nu \approx 0.1$ , tune  $M$ .

**Gradient boost (Alg 10.3):** init  $f_0$ ; for  $m=1..M$ : pseudo-resid  $r_{im} = -[\partial L / \partial f]_{f_{m-1}}$ ; fit tree to  $r_{im} \rightarrow$  regions  $R_{jm}$ ; leaf  $\gamma_{jm} = \arg \min_{\gamma} \sum_{R_{jm}} L(y, f_{m-1} + \gamma)$ ;  $f_m = f_{m-1} + \nu \sum_j \gamma_{jm} \mathbb{I}[x \in R_{jm}]$ . Sanity:  $L = \frac{1}{2}(y-f)^2 \Rightarrow r = y-f$ .

**AdaBoost.M1** ( $y \in \{\pm 1\}$ , NOT in ISLP): init  $w_i=1/N$ ; fit  $G_m$  weighted;  $\text{err}_m = \sum w_i \mathbb{I}[y_i \neq G_m] / \sum w_i$ ;  $\alpha_m = \log \frac{1-\text{err}_m}{\text{err}_m}$ ;  $w_i \leftarrow w_i e^{\alpha_m \mathbb{I}[y_i \neq G_m]}$ , renorm. Output  $\text{sign}(\sum \alpha_m G_m)$ .  $\text{err} < \frac{1}{2} \Rightarrow \alpha > 0$ . Equivalent to forward-stagewise additive mod. under  $L = e^{-yf}$ .

**Loss menu / -grad:** sq  $\rightarrow y-f$ ; abs  $\rightarrow \text{sign}(y-f)$ ; exp  $\rightarrow$  AdaBoost; binomial dev  $\rightarrow y-p$ ; multinomial  $\rightarrow \mathbb{I}-p_k$  fits  $K$  trees/round.

## Unsupervised

**Hierarchical by hand (-1/mistake):** (1) min off-diag  $D_{ij} \rightarrow$  merge at height  $D_{ij}$ . (2) Update: single = min (CHAINS), complete = max (BALANCED), average = mean. (3) Repeat. (4) Dendrogram:  $y$  = fusion height,  $x$  arbitrary (T/F: “close on  $x \Rightarrow$  similar” FALSE). **First merge identical** under all linkages. Ward/centroid OUT.

**PCA:** loadings  $\sum_j \phi_{jm}^2 = 1$ ; score  $z_{im} = \sum_j \phi_{jm} x_{ij}$ ;  $\text{Var}(Z_m) = \lambda_m$ ;  $\text{PVE}_m = \lambda_m / \sum_k \lambda_k$ . **Standardise first** (else max-var variable dominates—not scale-invariant). After standardise,  $\sum \lambda = p$  so  $\text{PVE} = \lambda/p$ . “90% recipe”: cumulate till  $\geq 0.90$ . “Most contributing var to PC1” =  $\arg \max_j |\phi_{1j}|$ . Loadings unique up to *sign*.

Scree (PVE per  $m$ ): elbow. Cumulative: threshold 90/95/99. Supervised (PCR): use CV, not elbow.

**K-means:**  $\min \sum_k \sum_{i \in C_k} \|x_i - \bar{x}_k\|^2$  (squared Euclidean). Algo: init  $\rightarrow$  centroid=mean / reassign-to-nearest until stable. Local min only  $\Rightarrow$  run **nstart**  $\geq 20$ , keep min WSS.  $K$  is *not* chosen by algo; elbow on WSS, domain. Standardise. K-means++/gap/silhouette OUT. Correlation dist  $d = 1 - \rho$  when *shape* matters (preferences) not magnitude.

## Neural networks

**Params, 1-hidden**  $p \rightarrow M \rightarrow C$ :  $M(p+1) + C(M+1)$  (bias on *receiving* layer). Multilayer: sum (prev+1)  $\cdot$  this. **Canonical wrong answer: forgetting biases.** Dropout doesn’t reduce param count.

**Why nonlinearity:** linear-only collapses to linreg. UAT:  $\geq 1$  hidden layer of squashing activation + enough width  $\Rightarrow$  approximates any Borel-meas  $f$ . *Existence, not construction; proof OUT.*

|               | $g(z)$                   | $g'(z)$             |
|---------------|--------------------------|---------------------|
| sigmoid       | $1/(1+e^{-z})$           | $\sigma(1-\sigma)$  |
| ReLU          | $\max(0, z)$             | $\mathbb{I}[z > 0]$ |
| softmax (out) | $e^{z_c} / \sum e^{z_j}$ | —                   |

**Output/loss pairing:** regression  $\rightarrow$  linear + MSE; binary  $\rightarrow$  sigmoid + BCE  $-\sum [y \log \hat{p} + (1-y) \log(1-\hat{p})]$ ; multi-class  $\rightarrow$  softmax + CCE  $-\sum y_c \log \hat{p}_c$ .

**SGD:**  $\theta \leftarrow \theta - \eta \hat{\nabla} L$ ,  $\hat{\nabla} L = \frac{1}{m} \sum_{i \in B} \nabla L_i$  (unbiased).  $\eta \leq 0.1$ ;  $\eta = 2$  “horrible.” Batch sizes powers of 2 (hardware). **Larger batch  $\Rightarrow$  less noise  $\Rightarrow$  less implicit  $L^2$ .**

**Regularisation MANDATORY** (iron rule, “never train without”). Menu: L1/L2 weight decay (excl. biases), mini-batch SGD (implicit L2), dropout, early stopping (**watch val loss, not train**), data augmentation, label smoothing, transfer learning.

**Dropout 0.2–0.5; never 0.5 in practice;** scale survivors  $1/(1-p)$  at train; no dropout at test.

Equiv to in-network ensembling.

**Implicit L2 (headline):** over-param regime, SGD  $\rightarrow$  min- $L^2$ -norm interpolator. Bridge to double descent.

**Backprop = chain rule reusing forward intermediates.** Seeds:  $\delta^{\text{out}} = -(y-f)$  for MSE;  $\delta_c^{\text{out}} = f_c - y_c$  for softmax+CCE;  $\delta^{\text{hid}} = (W^{(\ell+1)\top} \delta^{(\ell+1)}) \odot g'(v^{(\ell)})$ .

CNN one-liner: shared local filters + max-pool, conv $\rightarrow$ pool $\rightarrow$ flatten $\rightarrow$ dense $\rightarrow$ softmax; translation equivariance via weight sharing. RNN:  $A_t = \sigma(b + WX_t + UA_{t-1})$ ; weights shared across  $t$ . LSTM/GRU/BPTT/Adam/init/BN OUT.

**Notation drift** (open-book trap): prof  $\nu =$  ISLP  $\lambda$  (collides with ridge); prof  $M =$  ISLP  $B$ ; prof  $J =$  ISLP  $d+1$ .

## Cross-cutting (must-haves)

**Standardise before:** Ridge / Lasso / EN / PCR / PLS / PCA / K-means / hierarchical / KNN / NN. Compute mean,SD on *training*, reuse on test. Don’t standardise binary dummies. Intercept never penalised.

$K-1$  **dummies** for  $K$ -level factor; coefs = differences vs reference; sign-flip on ref-swap; predictions unchanged. Adding all  $K$  + intercept = rank-deficient.

**Hierarchical principle** (interactions): keep main effects whenever interaction is in the model.

**Generative vs discriminative.** Generative (LDA/QDA/NB): model  $f_k(x)\pi_k$ , flip via Bayes. Discriminative (logistic/KNN): model  $P(Y|X)$  directly.

$2 \times 2$  **direction rules (quick reference):**

flex/ $d/K$ -knots  $\uparrow$ : train $\downarrow$ , test $\uparrow$ , bias $\downarrow$ , var $\uparrow$

$\lambda$  (ridge/lasso/smoothing): bias $\uparrow$ , var $\downarrow$ , df $\downarrow$

KNN  $K \uparrow$ : smoother (bias $\uparrow$ , var $\downarrow$ ); LOESS span same direction

RF  $m \downarrow$ : trees more decorrelated (var $\downarrow$ )

Boosting  $\nu \downarrow \Rightarrow$  need  $M \uparrow$  (better gen.)

Batch size  $\uparrow$ : less SGD noise  $\Rightarrow$  less implicit  $L^2$

Threshold $\downarrow$ : sens $\uparrow$ , spec $\downarrow$

$\pi_k \uparrow$  (LDA): boundary moves *away* from class  $k$

Heterosc., correlated errors:  $\hat{\beta}$  unbiased, SE/CI wrong

PCA without standardising: max-var variable dominates PC1.

## Exam meta

4h open-book Inspira; ISLP PDF searchable; one A5 + calculator; *no code*, pseudocode/math/English all OK.

**Q types:** MC / T-F (write just T/F or short reason); output interpretation; hand calc (odds $\leftrightarrow p$ , df, MSE, PCA PVE, dendrogram); pseudocode; *one* mathy theory (bias-var, LDA boundary, or bagging-variance most likely).

**Method-compare template** (closing Q): quote both metrics  $\rightarrow$  direction of diff  $\rightarrow$  tie to bias-var / interpretability / FP-vs-FN cost  $\rightarrow$  pick + name criterion.

**Show work always; no negative marking; tricky Q  $\Rightarrow$  state your interpretation.** Past-exam code questions  $\rightarrow$  rewrite mentally as “interpret this output.” OUT: AIC/BIC algebra,  $F$ -test mechanics, VIF, SVM, survival, multi-testing, multi-class logistic, probit, history, optimizer internals, package names.